

Fixed-Start Cover Times Change Under Every Edge Addition on Connected Cyclic Graphs of Order Eight

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Abstract

Let G be a connected non-tree simple graph on eight vertices, let uv be a nonedge, and let s be any starting vertex. We prove, by an exhaustive exact computation, that adding uv changes the expected cover time of discrete-time simple random walk:

$$t_{\text{cov}}(G + uv, s) \neq t_{\text{cov}}(G, s).$$

The census contains all 11,094 unlabelled connected cyclic graphs of order eight, all 150,573 nonedges, and all 1,204,584 marked graph–nonedge–start instances. Two exact formulations are used: a visited-set recurrence and target-set inclusion–exclusion with absorbing hitting-time systems. Their computations use different primes; every encountered grounded Laplacian is nonsingular modulo the relevant prime, and neither computation finds a zero difference residue. Inventory regeneration, rooted canonicalization, exact-rational regression, and a clean full rerun provide additional checks. This is only the connected non-tree, order-eight, fixed-start layer; it makes no assertion about worst-start cover time, higher orders, or the unrestricted strict-nonequality conjecture.

1 Statement and context

For a finite connected simple graph H and discrete-time simple random walk $(X_t)_{t \geq 0}$ started at s , set

$$\tau_{\text{cov}} = \min\{t \geq 0 : \{X_0, \dots, X_t\} = V(H)\}, \quad t_{\text{cov}}(H, s) = \mathbb{E}_s \tau_{\text{cov}}.$$

Thus the start is visited at time zero, and time is counted in edge transitions. Cover times are classical objects in random-walk theory [1, 3].

Si studied the exact response to adding a nonedge and conjectured strict fixed-start nonequality for every finite connected simple graph [5, Conjecture 6.2]. The exact census in [5, Theorem 6.4] excludes fixed-start equality through order seven and also among trees of orders eight and nine. The theorem below settles exactly the complementary connected non-tree order-eight layer. This comparison is a statement about the scope of the cited census, not a claim of priority beyond the literature examined here.

The certification style—an explicitly delimited finite theorem, exhaustive isomorphism-class enumeration, exact predicates, and a structurally separate check—also appears in finite classifications in discrete geometry and graph spectra [7, 6]. Those works are methodological comparisons only and support no cover-time assertion in this article.

Theorem 1 (Order-eight fixed-start strictness). *Let G be a connected non-tree simple graph on exactly eight vertices. For every unordered genuine nonedge $\{u, v\}$ of G and every $s \in V(G)$,*

$$t_{\text{cov}}(G + uv, s) \neq t_{\text{cov}}(G, s).$$

No sign is asserted in Theorem 1; the difference may be positive or negative. The start may be either endpoint of the added edge.

2 Exact algebraic formulations

Let $H = (V, E)$ be connected, with degree matrix D and adjacency matrix A_H . For a nonempty proper subset $B \subsetneq V$, define the grounded principal matrix

$$M_B = D[B, B] - A_H[B, B].$$

Lemma 2 (Grounded nonsingularity). *Every M_B is positive definite over $\mathbb{R}$, and hence nonsingular over $\mathbb{Q}$.*

Proof. For $z \in \mathbb{R}^B$,

$$z^\top M_B z = \sum_{xy \in E(H[B])} (z_x - z_y)^2 + \sum_{\substack{xy \in E(H) \\ x \in B, y \notin B}} z_x^2.$$

Every component of $H[B]$ has an edge to $V \setminus B$, since H is connected and B is proper. If the displayed form vanishes, z is constant on each component and is zero at a boundary vertex of each component. Therefore $z = 0$. $\square$

2.1 Visited-set recurrence

For $x \in B \subseteq V$, let $C_H(x, B)$ denote the expected additional time needed to cover H when the walk is at x and precisely the set B is already marked visited. The boundary values are $C_H(x, V) = 0$. Multiplying the one-step recurrence by $d_H(x)$ gives, for nonempty proper B ,

$$d_H(x)C_H(x, B) - \sum_{\substack{y \in B \\ y \sim_H x}} C_H(y, B) = d_H(x) + \sum_{\substack{y \notin B \\ y \sim_H x}} C_H(y, B \cup \{y\}). \quad (1)$$

The coefficient matrix is M_B , and the right-hand side is already known when subsets are processed in decreasing cardinality. Lemma 2 therefore makes the recursion uniquely solvable over $\mathbb{Q}$, with

$$t_{\text{cov}}(H, s) = C_H(s, \{s\}).$$

We call this Method A.

2.2 Absorbing hitting times and inclusion–exclusion

For nonempty $S \subseteq V$, let T_S be the first hitting time of S , and write $h_S(x) = \mathbb{E}_x T_S$. On $U = V \setminus S$, the absorbing equations are

$$M_U h_S = d_U, \quad (2)$$

where d_U is the vector of degrees in H indexed by U ; also $h_S = 0$ on S . If T_v is the first hitting time of vertex v , then $\tau_{\text{cov}} = \max_{v \in V} T_v$. Applying inclusion–exclusion to the tail event that some vertex has not yet been hit and summing over time yields

$$t_{\text{cov}}(H, s) = \sum_{\emptyset \neq S \subseteq V} (-1)^{|S|+1} h_S(s). \quad (3)$$

Indeed, an intersection of the events $\{T_v > t\}$ over $v \in S$ is $\{T_S > t\}$. Equations (2)–(3) form Method B. It uses absorbing hitting-time systems and inclusion–exclusion rather than visited-set states.

2.3 Finite-field exactness

Lemma 3 (Good-prime certificate). *Fix a prime p . If every grounded matrix encountered by one of the two methods is nonsingular over $\mathbb{F}_p$, then its finite-field output is the reduction of its unique rational output. Consequently, for graphs H_1, H_2 on the same marked vertex set,*

$$t_{\text{cov}}(H_1, s) = t_{\text{cov}}(H_2, s) \implies t_{\text{cov}}(H_1, s) \equiv t_{\text{cov}}(H_2, s) \pmod{p}.$$

Thus a nonzero difference at a good prime proves rational nonequality.

Proof. Each matrix has integer entries. Nonsingularity modulo p says that its determinant is not divisible by p , so inversion commutes with reduction of the corresponding rational system. This proves the claim directly for Method B. Descending induction over $|B|$ in Equation (1) proves it for Method A. Equality over $\mathbb{Q}$ would then force equality after reduction. $\square$

No rational reconstruction or floating-point tolerance is involved in this argument.

3 Exhaustive census and proof

The unlabelled inventory was generated by `geng` from `nauty/Traces` [4] using

`geng -cq 8 8:28.`

It contains one representative of every connected simple graph on eight vertices with between 8 and 28 edges. A connected order-eight graph is a non-tree exactly when it has at least eight edges. The stream has 11,094 distinct graph6 records. The complete graph is present but contributes no nonedge.

For each representative, the decisive loop inspected all 28 unordered vertex pairs, skipped precisely the existing edges, and then inspected all eight starts. No nonedge or start orbit reduction was used. The principal counts and outcomes are shown in Table 1.

Table 1: Exact census totals. The final pair lists Method A at $p = 1,000,003$ and Method B at $p = 1,000,033$.

Quantity	Value
Connected non-tree graph representatives	11,094
Genuine nonedges	150,573
Marked graph–nonedge–start instances	1,204,584
Base graph/start/prime cross-method comparisons	177,504
Direct augmented graphs solved by each designated method	150,573
Direct marked comparisons by each designated method	1,204,584
Singular principal systems	0
Zero difference residues, Methods A/B	0/0

For every base graph, both methods computed all eight starts at both primes and agreed in all $11,094 \cdot 8 \cdot 2 = 177,504$ comparisons. For every nonedge, the augmented adjacency matrix was constructed directly. Method A then compared all eight augmented values with their base values modulo 1,000,003, while Method B did so independently modulo 1,000,033. The implementation tests both moduli for primality, uses pivoting, and fails closed if any pivot is absent. No such failure occurred, and each method reported zero equal residues among all 1,204,584 direct comparisons.

Proof of Theorem 1. The inventory contains one representative of every unlabelled graph in the quantified class. Fix a representative, a nonedge, and a start. Each designated computation used a good prime and returned a nonzero difference. Lemma 3 therefore proves that the corresponding rational cover times are unequal. This was done for every one of the 1,204,584 marked instances.

Finally, any labelled graph in the class is isomorphic to an inventory representative. Transporting its nonedge and start gives a marked instance processed above, and a rooted graph isomorphism preserves the random walk and the cover event. The conclusion therefore holds for every labelled graph as well. $\square$

4 Independent checks and reproducibility

The following checks supplement the direct proof computation.

1. An independent NetworkX [2] decoder validated all 11,094 graph6 records, connectivity, distinctness, edge counts, and graph6 round trips. A fresh unrestricted connected order-eight generation produced 11,117 classes: 23 trees with seven edges and the required 11,094 graphs with at least eight edges.
2. Every rooted base and augmented instance was separately canonicalized with `labelg`, placing the start in a singleton color class. Lookup through the resulting 72,374 rooted isomorphism classes again gave no zero difference residue at either prime. This lookup is a check, not an input to the decisive direct scan.
3. Exact-rational regression compared the two formulations on three order-seven graphs and their augmentations, agreeing on all 42 cover-time values. Four order-eight base/augmentation pairs, spread across edge counts 8, 8, 14, 27, were subsequently recomputed over $\mathbb{Q}$; the formulations agreed for every start and every checked difference was nonzero.
4. A clean reproduction regenerated the graph inventory and both rooted streams, recompiled the census program, reran the entire marked census and the rational regression, and reproduced every non-timing output field.

The accompanying computational supplement contains `census_modular.cpp`, the graph6 inventory, the rooted streams, `exact_regression.py`, `graph6_crosscheck.py`, their machine-readable results, and `reproduce.sh`. The recorded build used Apple clang 21.0.0, nauty 2.9.3, NetworkX 3.6.1, and Python 3.12.14. The core census can be rebuilt and run as

```
clang++ -std=c++20 -O3 -DNDEBUG -Wall -Wextra -pedantic \
    census_modular.cpp -o census_modular
./census_modular order8-connected-nontrees.g6 \
    base-rooted-canonical.g6 augmented-rooted-canonical.g6
```

The reproduction script also regenerates these inputs with the recorded nauty tools before comparison.

5 Scope and assistance disclosure

Theorem 1 is a finite, computer-assisted theorem. It does not address trees, graphs of another order, edge conductances other than one, the sign of the change, or the maximum over starting vertices. In particular it does not settle the unrestricted conjecture in [5]. Its exactness comes from Lemma 3 and the exhaustive certificate, not from a numerical tolerance or a statistical test.

Exact enumeration, finite-field linear algebra, rational regression, graph6 validation, and reproduction were performed by the disclosed programs. An AI language model assisted with organizing the exposition and checking it against the accepted computational records. AI output is not treated as mathematical evidence; the theorem-critical claims are supported by the stated argument and replayable exact artifacts.

References

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